EXECUTIVE SUMMARY

Representation Geometry, Latent Segmentation and Analytic Aggregate Risk in Motor Insurance

This note documents a computational experiment in motor insurance pricing and aggregate risk measurement using the freMTPL2 benchmark (678,013 policy-years). The project investigates how representation geometry propagates through the actuarial pipeline — from latent segmentation and pricing discrimination to aggregate capital metrics computed analytically through FFT-based collective-risk methods.

The starting point was practical: replace Monte Carlo aggregate simulation with deterministic FFT convolution, while testing whether latent segmentation can simultaneously improve pricing discrimination and aggregate portfolio behaviour.

Three representation families are compared:

- A-series: classical one-hot categorical encoding,
- B-series: Burt-replacing latent geometry (categorical dummies replaced by Burt coordinates),
- and C-series: Burt-augmented geometry (categorical dummies retained alongside Burt coordinates).

The Burt representation is constructed from the Burt matrix $B = Z^T Z$, embedding categorical modalities into a continuous co-occurrence geometry related to Multiple Correspondence Analysis (MCA). Gaussian Mixture Models (GMMs) are then fitted on actuarially enriched feature spaces combining:

- categorical and numerical features,
- predicted claim frequency,
- and predicted claim severity.

Several findings emerged.

1. Soft probabilistic segmentation materially outperforms hard clustering. Enriched GMM improves pricing discrimination by approximately 5.5 Gini points relative to the plain Tweedie GLM baseline, while enriched K-means on the classical geometry actually falls below the plain GLM baseline by 0.9 Gini points — a negative result confirming that hard clustering on an enriched feature space can actively harm discrimination when soft probabilistic assignment is unavailable.

2. The Burt representation reveals an unexpected trade-off between discrimination and calibration. Relative to the best classical enriched geometry, the Burt-based models

sacrifice approximately 4.5 Gini points but improve top-decile calibration by roughly 10 percentage points.

3. Following the recommendation by Hainaut & Denuit (Detra Note 2025-2), Tweedie deviance per exposure is added as a second fit-quality metric alongside the Gini coefficient. A5 achieves the lowest deviance (61.276 per exposure unit) across all 21 experiments — the same experiment that wins on Gini. Within the A-series, deviance and Gini improve together as segmentation quality increases: the two metrics are aligned. However, the cross-series correlation is weak and not statistically significant. The B and C-series cluster at competitive deviance values despite substantially lower Gini, suggesting that the Burt representation fits the loss distribution well in absolute terms while redistributing risk ranks differently from classical encoding. The top-decile calibration ratio is a genuinely orthogonal dimension that neither Gini nor deviance captures.

4. Interestingly, retaining both the original categorical factors and the Burt coordinates (the augmented representation) produces results identical to the Burt-only representation to four decimal places of Gini, confirming informational completeness. The original categorical factors carry no discriminating signal beyond what the Burt coordinates already encode. The C-series is therefore excluded from the canonical comparison, which is between the best classical GMM and the best Burt-replacing GMM, holding clustering method constant.

5. The aggregate distributions are evaluated analytically via compound Poisson FFT convolution rather than Monte Carlo simulation, producing exact and reproducible VaR and TVaR figures. A key modelling input is the severity coefficient of variation per segment, which is estimated empirically from observed claim amounts using weighted lognormal MLE — with GMM soft membership weights restricted to claimants. The empirical CVs range from approximately 1.1 to 1.8 across segments: TVaR 99% increases by approximately €500,000 relative to an original fixed-CV assumption of 0.8. This confirms that segment-level severity heterogeneity is both estimable from data and consequential for aggregate tail measurement.

6. A separate severity-screening exercise shows that Burr XII and log-logistic distributions fit the empirical claim severity substantially better than lognormal under AIC criteria. However, the fitted Burr XII parameters imply near-infinite higher moments, leading to numerical instability within the FFT aggregation framework – a disappointment. Lognormal severity is therefore retained not because it optimizes local fit, but because it preserves tractable aggregate moment behaviour.

7. The coverage-layer analysis reveals that the preferred representation depends on the reinsurance structure. On a ground-up basis, the Burt-based model (B5) produces a TVaR 99% approximately €85,000 lower than the classical model (A5). Under a

deductible structure, this advantage widens to approximately €550,000. Under a per-claim limit, the ranking reverses in favour of the classical geometry, with A5 producing a TVaR 99% approximately €478,000 lower than B5. This suggests that representation geometry interacts with treaty structure in ways that are not visible through discrimination metrics alone — the optimal choice of representation depends not only on the pricing objective but on the reinsurance programme in place.

Overall, the experiments suggest that representation engineering influences not only pricing performance, but also aggregate portfolio behaviour, numerical tractability, and capital measurement. The strongest pricing discrimination remains associated with supervised actuarial enrichment, while relational categorical geometry appears to moderate certain aggregate tail characteristics within the FFT collective-risk framework.

These findings are established under linear Tweedie pricing. A gradient boosting extension has since tested their stability under a non-linear learner; see the *Gradient Boosting Extension* (June 2026).